MITx 6.419x Data Analysis: Statistical Modeling and Computation in Applications

Networks

PhilMerlin, 2021-04-21

Problem 1 – Suggesting Similar Paper

Part (c): (2 points)

Let

- n be the number of papers in the network *(the number of papers under study)*.
- A be the adjacency matrix of the citation network

We have already shown that we can write the co-citation weighted adjacency matrix as $C = A^T \cdot A$.
A is of size n x n. Hence **time complexity for matrix multiplication is $O(n^3)$, same as our friend's algorithm.**

Please note that more optimized matrix calculation can lead to lower time complexity: "As of December 2020, the matrix multiplication algorithm with best asymptotic complexity runs in $O(n^{2.3728596})$ time, given by Josh Alman and Virginia Vassilevska Williams" – Wikipedia

Part (d): (3 points)

Bibliographical coupling:

- Is **retrospective** / looking backwards: two papers will be related if they share same "ancestors";
- Is **static** / frozen in time: new progresses in a field won't modify coupling;
- May not capture the right relationship: **different pieces of information might be referred to** by the two coupled papers;

Co-citation:

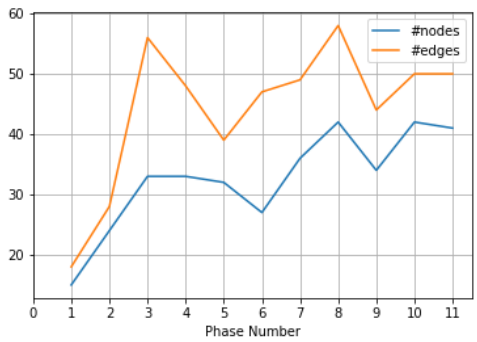
- Is adaptative / **evolving throughout time** (and the number of links will tend to increase, contrarily to bibliographical coupling);
- **Aggregates the views of many different authors** (also less prone to timing or "fashion" effect – recent articles might be more prevalent in bibliographies);

Thus, they can give noticeably different results. Imagine the advent of two new fields based on an ancestor field. Say for illustration that biostatistics and financial engineering are just being created out of statistics. A foundation paper in each of these two fields would cite a number of well-known statistics papers. Bibliographical coupling between these two papers would be high, although they relate to two totally different domains. It would stay high over time. At inception, co-citation would be null, and would remain very low while these two fields develop. Co-citation clearly performs much better in this caricatural case.

Hence, **co-citation makes a better measure of similarity between papers**.

Problem 2 – Investigating a time-varying criminal network

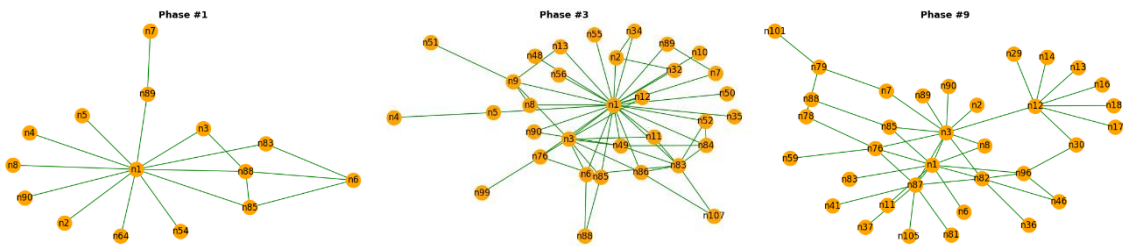
Part (c): (2 points)



The number of nodes increases sharply over the first few phases as **the organised network is in the process of being created**:

- In phase 1, **n1** (Daniel Serero, the "godfather") is probably contacting his connections to launch the trafficking network. One of them is **n83** (Alain Levy, investor and transporter of money). *See drawn networks below.*
- In phase 3, we observe that the originally centralized network has become a kind of spider web: **peripheral connections** get connected to each other, and **bring their connections to the network. Hence the increasing number of nodes.** For instance, **n83** probably brought in **n86** (Gerard Levy, presumably relative or even maybe his brother *(unchecked)*), **n107**, **n84** and **n52**.

After the initial phases, **all players are in the game have organised themselves and consequently the number of nodes levels out.**



Based on this observation, **we should adjust our approach** in Part (b) Question 5:

- In order to detect the instigators or the criminal network (and not necessarily the operating soldiers), we could **appose weights on the first few phases** to get the most central nodes (for instance weights decreasing phase after phase).

Here, equally weighting phases 1 to 3 (weight 1/3 for phases 1, 2 and 3, weight 0 for all the following phases) gives:

	Degree Centrality	Betweenness Centrality	Eigenvector Centrality
Highest	n1	n1	n1
2nd highest	n3	n89	n3
3rd highest	n83	n87	n83

Resulting in **top 3 key players**:

	Top 3 Central Players – Average Centrality Ranking - Phase 1 to 3	Person : role *(with hindsight)*
Highest	**n1**	Daniel Serero : Mastermind of the network
2nd highest	**n3**	Pierre Perlini : Principal lieutenant of Serero
3rd highest	**n83**	Alain Levy : Investor and transporter of money

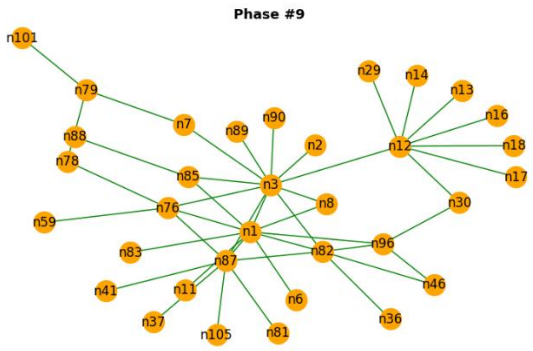
Part (d): (5 points)

Degree centrality

- It identifies the individuals with the **highest number of direct connections** (the nodes with the highest number of edges).
- It reveals the most connected players, but a **limitation** is that **it doesn't necessarily lead to identifying the mastermind of the network: the most connected player might be, at a given phase, a middle man with operational responsibilities, for instance organizing a specific operation**. For instance, in phase 11, **n12** (Ernesto Morales, principal organizer of the cocaine import) has the highest degree centrality. He is obviously a key player, but probably not the first one police would like to catch.

Betweenness centrality

- It identifies the **individuals through which most information flows**. In other words, arresting them could disrupt the criminal network most, as the transmission of orders and decisions (and maybe drug itself) would be slowed down most.
- Although it succeeds in most phases at identifying n1 as the most important player, it fails to do it consistently (the mastermind might judge better to hide for a while after his instructions went transmitted). This is the case in phase 9, where **n3** (Pierre Perlini, principal lieutenant of Serero) and **n12** (Ernesto Morales, principal organizer of the cocaine import) are recognised as the most central nodes (using betweenness measurement).

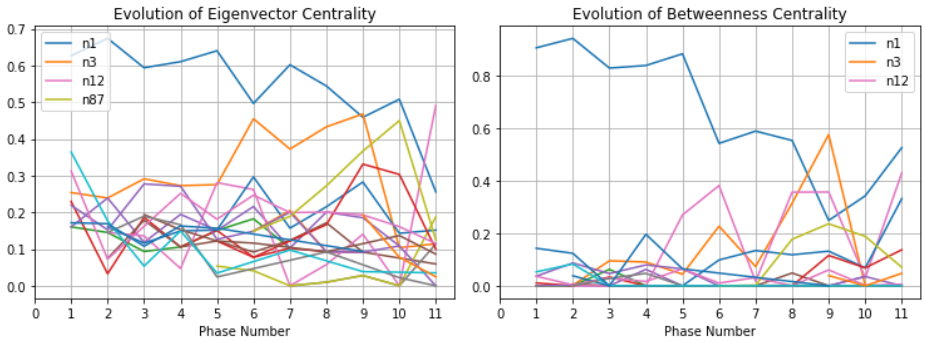


Eigenvector centrality

- It **identifies the most influential connections**, as nodes get weighted according to the importance of the neighbouring connections.
- As such, it is the **best measure to identify who is running the illegal activities**. The individuals with the most influence over the network will be spotted.

In practice though one can argue between Eigenvector (left) and Betweenness (right) centrality. Possible ways to draw more clear-cut conclusions would be to:

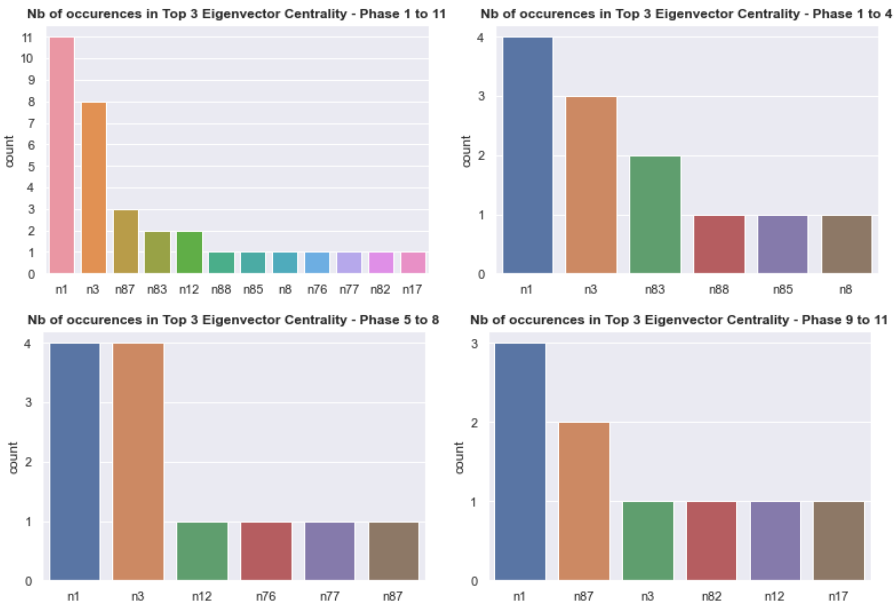
- Either weight centrality with phase (decreasing weights) and look at a weighted centrality across all phases;
- Or possible use a directed network instead of the undirected one we've worked on so far.



Part (e): (5 points)

As already hinted above, we'll **define the importance of an actor's role as his influence over the network**. As a consequence, the best centrality measure that identifies the most important actors is **eigenvector centrality**.

Counting the number of times each player appears among the most central ones (using eigenvector centrality) gives the following hindsight:



Resulting in **most important actors** *(with varying importance across time as seen above)*:

Rank	Most important actors	Person : role *(with hindsight)*
1	**n1**	Daniel Serero : Mastermind of the network
2	**n3**	Pierre Perlini : Principal lieutenant of Serero
3	**n87**	Patrick Lee : Investor
4	**n83**	Alain Levy : Investor and transporter of money
5	**n12**	Ernesto Morales : Principal organizer of the cocaine import

Part (f): (3 points)

At phase 4, the first police intervention disrupted the network: a $2,500,000 seizure of marijuana took place.

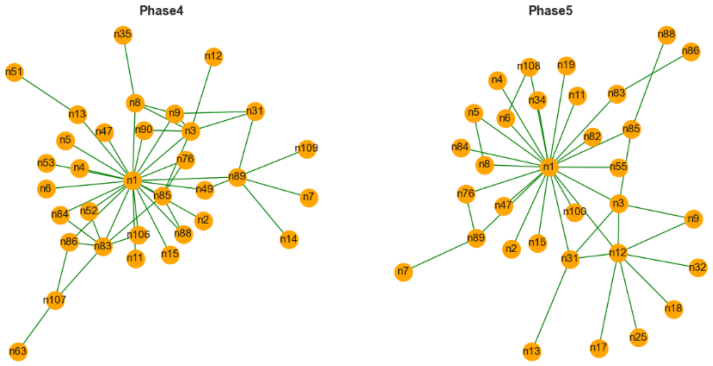
The different centrality measures reveal a noticeable phenomenon: **n12** (Ernesto Morales, principal organizer of the cocaine import) **increases dramatically in the rankings up to the 2nd position**. Increase is also apparent for **n82** (Salvatore Panetta, transport arrangements manager) and **n17** (Oscar Nieri, the handyman of Morales).

	Degree Centrality		Betweenness Centrali		Eigenvector Centrality	
Ranking	Phase 4	Phase 5	Phase 4	Phase 5	Phase 4	Phase 5
1	1	1	1	1	1	1
2	83	12	89	12	3	12
3	3	3	3	89	83	3
4	89	89	83	85	85	85
5	85	85	8	83	8	89
6	8	76	86	3	86	76
7	86	83	85	82	89	8
8	106	8	106	17	106	6
9	84	6	84	84	84	5
10	76	5	76	86	76	83
11	88	82	12	76	88	82
12	12	17	11	11	11	84
13	11	84	88	88	6	11
14	6	86	6	8	5	17
15	5	11	5	6	12	88
16	17	88	17	5	17	86
17	82	106	82	106	82	106
18	77	77	77	77	77	77
19	87	87	87	87	87	87
20	16	16	16	16	16	16
21	33	33	33	33	33	33
22	80	80	80	80	80	80
23	96	96	96	96	96	96

** Numbers corresponds to players (1 for n1...)*

	Ranking Change between Phase 4 and 5			
Player	Degree	Betweenness	Eigenvector	Average
n1	0	0	0	0
n3	0	-3	-1	-1
n5	5	-1	5	3
n6	5	-1	5	3
n8	-2	-9	-2	-4
n11	-2	0	-1	-1
n12	10	9	13	11
n16	0	0	0	0
n17	4	8	2	5
n33	0	0	0	0
n76	4	-1	4	2
n77	0	0	0	0
n80	0	0	0	0
n82	6	10	6	7
n83	-5	-1	-7	-4
n84	-4	0	-3	-2
n85	0	3	0	1
n86	-7	-4	-10	-7
n87	0	0	0	0
n88	-5	0	-4	-3
n89	0	-1	2	0
n96	0	0	0	0
n106	-9	-9	-9	-9

The network seems to diversify its illegal products with the **addition of a drug that is maybe less visible or cumbersome to carry, cocaine.** The new importance of n12 is visually obvious on the network's graphs:

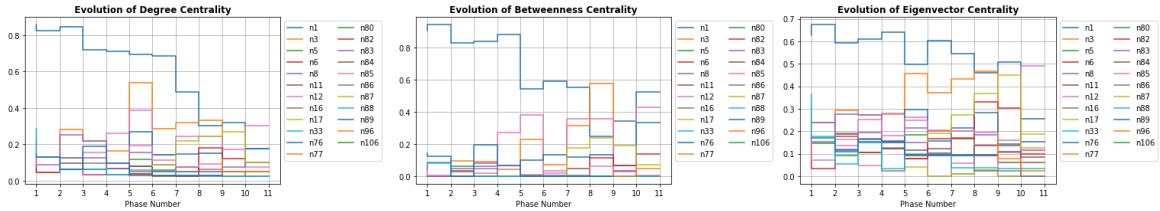


Mere coincidence or the result of the seizure of marijuana?

Part (g): (4 points)

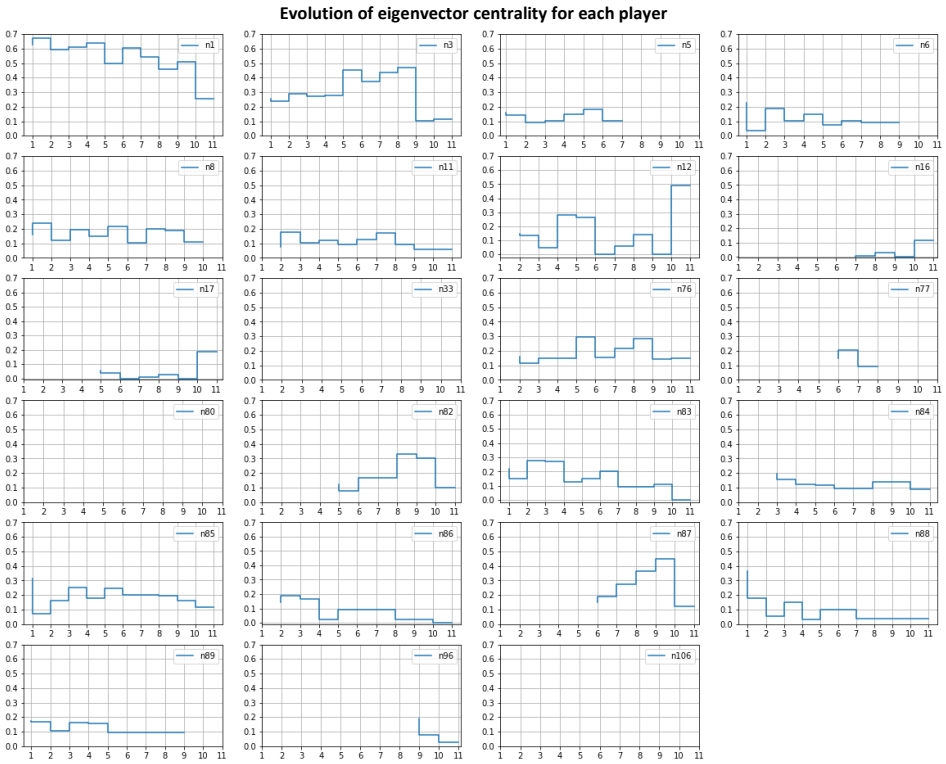
The coarse patterns below show:

- **n1** concentrates most of the centrality of the network until phase 4 and its importance gradually diminishes over time thereafter;
- **The centrality of the network, at first highly concentrated** (most of it in the hands of a single player) **evolves to a more spread-out structure**; From phase 8 onwards, a handful of key players share a relatively lower centrality level;



The evolution of eigenvector centrality of each player shows:

- The roles of **n1** and **n3** (Daniel Serero and Pierre Perlini, his principal lieutnant) tend to swap between phase 1 and 9. Both drop dramatically around phase 10.
- Around the same time, the roles of **n12** and **n17** (Ernesto Morales and Oscar Nieri, his handyman) increase sharply.

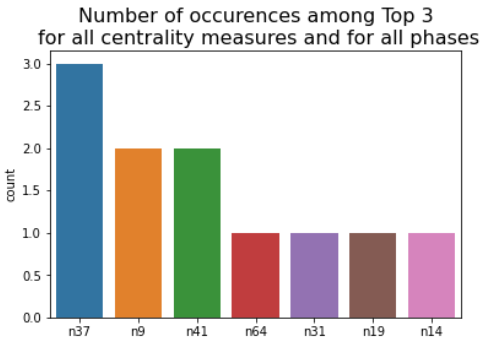


These observations show:

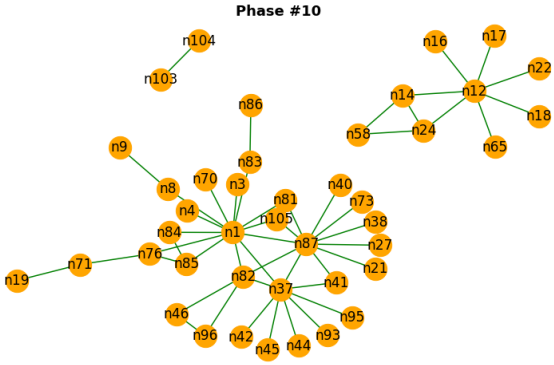
- **A progressive decentralization of influence** in the criminal network
- **A shift of "power" (loose extension of eigenvector centrality) from Daniel Serero and his initial marijuana traffic to Ernesto Morales and cocaine trafficking**.

Part (h): (2 points)

In order to analyse the important actors not listed in the investigation, let's firstly extract the unlisted ones that appear at least once among the top 3 centrality measures.



As can be seen for **n37** in phase 10 below, **these nodes connect the core of the network with other more peripheral nodes** (presumably dealers). They could be for instance 'wholesale' resellers breaking drug into smaller sellable quantities. Hence their centrality importance.



Part (i): (2 points)

In a criminal network such as a mafia organisation, the link between the participants is directional: the 'godfather' instructs his lieutenants, who instructs other middlemen, who bring instructions down in the organisation until each dealer receives his instructions. Therefore, **looking at the directed version of the criminal network would bring further information on influence and its direction**.

For each player in the network, **out-degree centrality characterises his initiative of contacting others** (sending instructions or information), while **in-degree centrality characterises the extent to which a player is being contacted** (receiving instructions or information).

In this exact vein, **left-eigenvector centrality measures the extent to which a player is contacting other important players**. **Right-eigenvector centrality measures the effect of being contacted by important players**.

Consequently, we would expect that masterminds show low left-eigenvector centrality and large right-vector centrality (and the opposite for street dealers).

Part (j): (4 points)

Initially used for analysing web pages, hubs and authorities provide following information on a directed network:

- **Hub: a node that points to many other nodes** *(a web page that points to many others)*
- **Authority: a node linked by many different hubs** *(a web page referenced by many others)*

Hubs and Authorities

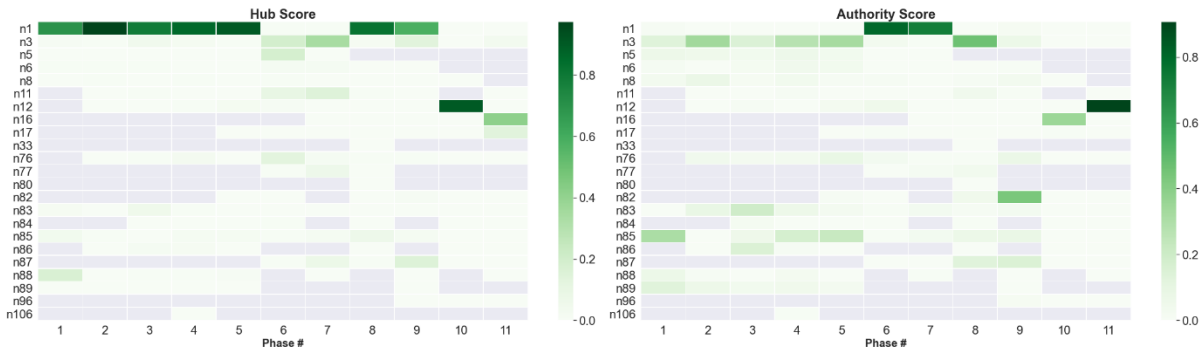
We now define two interdependent notions of centrality and combine them in a mutual recursion. An important **hub** is a node that points to many important **authorities** . An important authority is one that is pointed to by many important hubs.

We begin with an initial assignment of hub and authority scores for every node $\mathbf{x}^0$ and $(\mathbf{y}^0)^T$, respectively. The updates are as follows:

$$\mathbf{x}^{k+1} = \alpha A \mathbf{y}^k, \quad (\mathbf{y}^{k+1})^T = \beta (\mathbf{x}^{k+1})^T A.$$

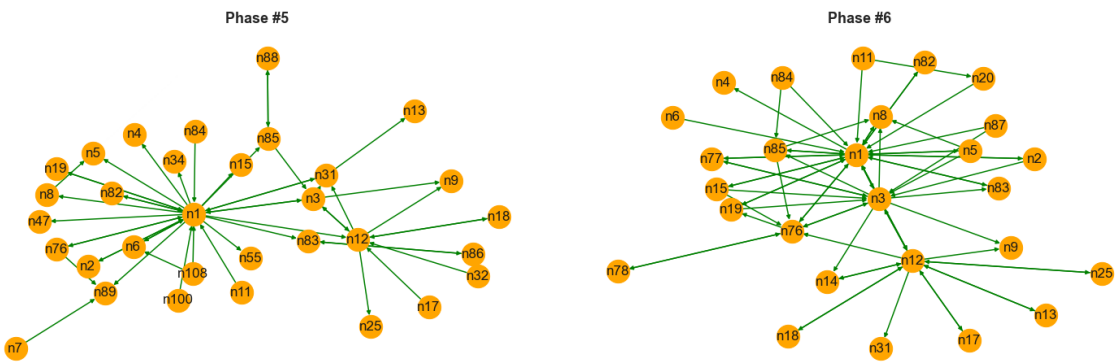
Choosing $\alpha\beta = 1/\lambda_{\max}(AA^T)$, we can show that $\mathbf{x}^k \to \mathbf{v}$ and $(\mathbf{y}^k)^T \to \mathbf{w}^T$, where $AA^T\mathbf{v} = \lambda_{\max}(AA^T)\,\mathbf{v}$ and $A^TA\mathbf{w} = \lambda_{\max}(A^TA)\,\mathbf{w}$. In fact, the non-zero eigenvalues of AA^T and A^TA are the same and $\mathbf{w} = A^T\mathbf{v}$.

Below are represented the normalized hub and authority scores for each phase and each player (among the 23 players under investigation).



From phase 1 to 5, and 8 to 9, **n1 acts as the most predominant hub** (remember the very centralized network observed at phase 1). During these phases, **n3 (n1's lieutenant) acts as a prominent authority.** We interpret this as **n1** possibly giving instruction to **n3**.

Phase 6 and 7, the roles of **n1** and **n3** are inverted. We could interpret this as a period of information feedback (presumably debrief on operations).



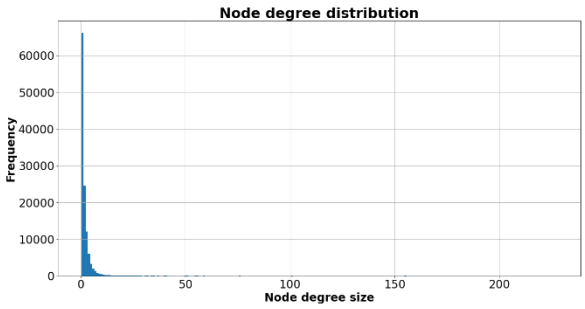
Although previous analysis in part (g) showed the key roles of these two players, **hub and authorities analysis gives further understanding of the information flow**.

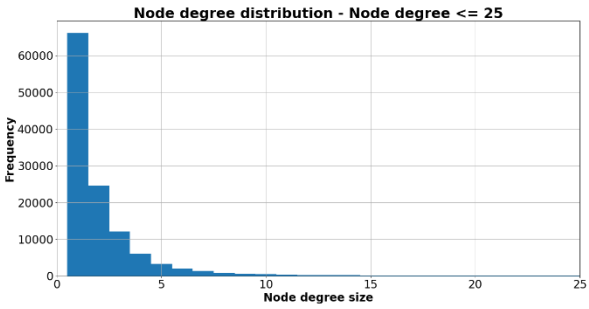
The same conclusion on the benefits of hubs and authority over undirected centrality is also valid for **n12**. We can interpret its hub and authority score in link with **n16** (Johnny Pacheco, cocaine provider): **n12** might give instructions to **n16** in phase 10 and receive operations feedback in phase 11.

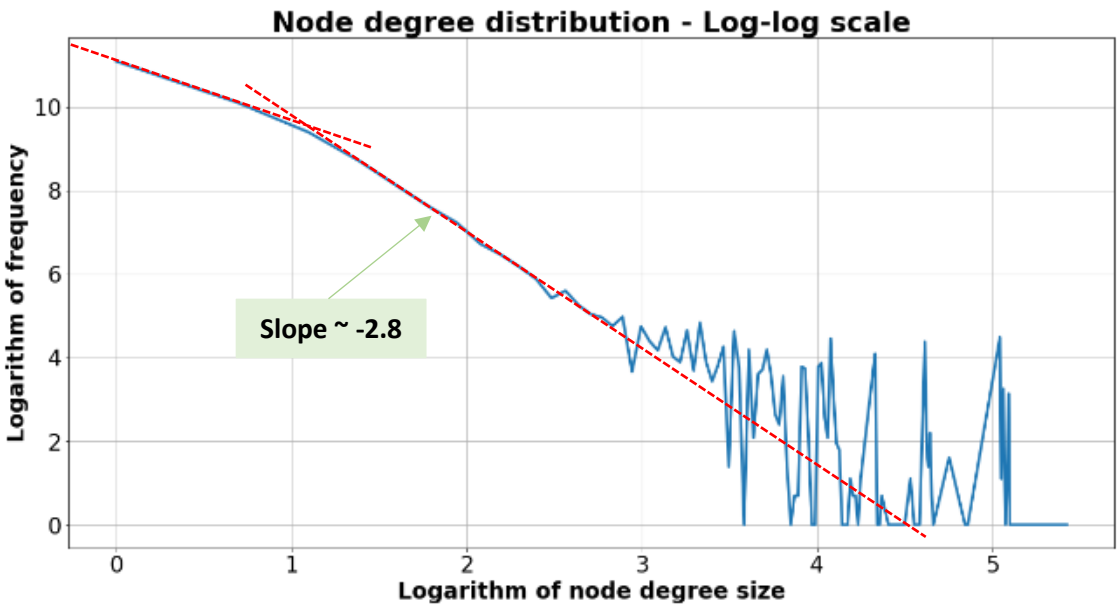
Problem 4 – Co-offending Network

Part (g): (3 points)

The degree distribution of the criminal network is plotted below, for different scaling including log-log scale.







The log-log plot shows a slight curvature at k = 3, k corresponding to node degree size. Consequently, **the degree distribution doesn't stricto sensu follow a power law distribution**. Nevertheless, fitting a power law distribution of slope -2.8 from k=3 onwards seems reasonable.

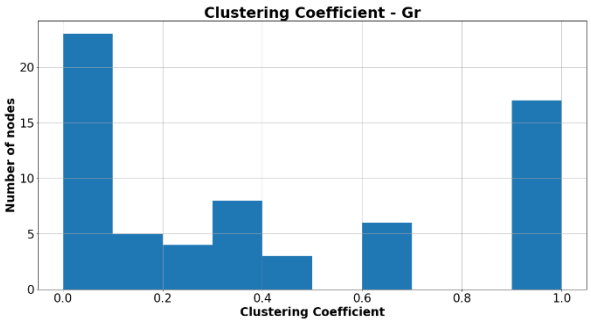
As such, **this graph couldn't have come from an Erdös-Renyi model**, as this one has a binomial degree distribution that falls off quickly (probability of observing high k is tiny).

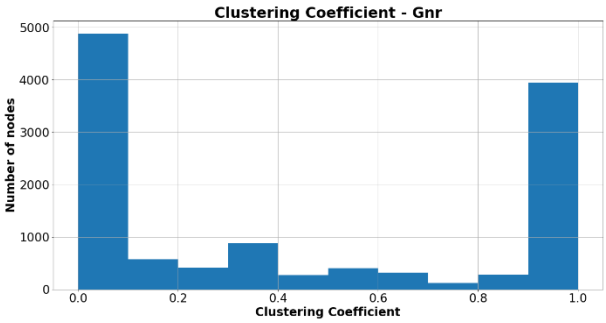
The different regimes {k < 3, k >=3} could be interpreted as follows:

- **The crimes committed by single or pairs of offenders might differ from the crimes committed by groups of more than 2 people**.

Part (m): (4 points)

The distributions of the clustering coefficients ("C") of Gr (criminals who co-offended together at least twice) and Gnr (criminals who never repeated a crime with the same individual) are shown below.





Both distributions are strongly weighted on the wings. Most of nodes have:

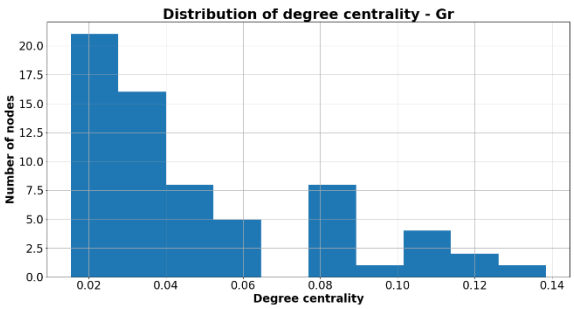
- **Either 0 clustering coefficient (no connections in their neighbourhood)**
- **or clustering coefficient close to 1 (well interconnected neighbours)**

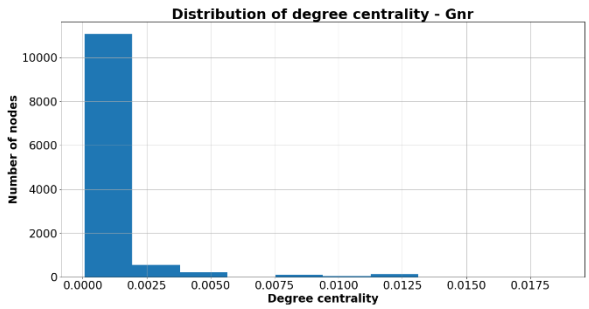
For Gr, this means that most of the criminals who co-offended repeatedly were either isolated to some extent (say pairs of criminals) or part of a larger and more connected structure (possibly a criminal ring). We could have expected heavier weights on intermediate C values to reflect middle-size criminal sub-networks.

Regarding Gnr, the most surprising is to observe that a large proportion of nodes is part of a well interconnected "community". Those guys know each other but never commit crimes together. Is this a feature of organised crime: never repeating an offense with the same partner in crime? *(is Hollywood a lie?).*

Part (n): (4 points)

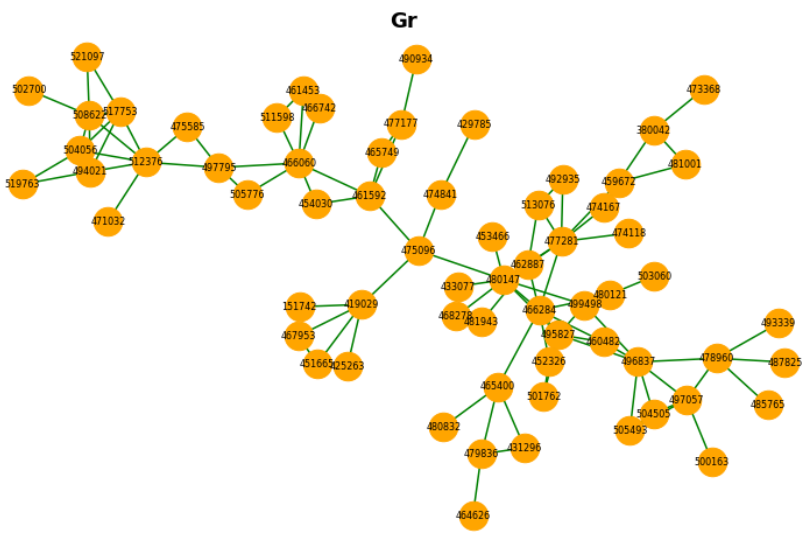
Let's analyse the degree centrality of these two graphs.





As we would expect, a **higher proportion of nodes is central in Gr compared to Gnr**. The most central actor in Gr (OffenderIdentifier==480147) committed 46 crimes.

This strengthens our intuition that **Gr could be a criminal ring**.



Project – Co-offending Network

Question of interest:
Is there organized crime related to prostitution in the data? How does this network compare with the organized crime in the CAVIAR network?

1. **Methodology**
   We'll approach this question with the following steps:
   a. **Data wrangling (collection and cleaning)**
      i. Extract a specific list of crimes
      ii. Construct the network
      iii. Remove isolated nodes
      They don't give any information on connected organizations.
   b. **Analysis**
      i. Network structure
         - Basic statistics
         - Degree distribution
         - Visualization
         - Clustering coefficient
      ii. Focus on largest component *(For simplicity's sake in this report)*
         - Visualization
         - Centrality measures *(In order to extract key players)*
            - Degree centrality
            - Betweenness centrality
            - Eigenvector centrality

   Both visual inspection of the network and centrality measures will serve as a basis of comparison with the CAVIAR drug ring.

2. Results

Network structure

We have filtered, out of the database of arrestations in Quebec, the following crime types:

Value	Label
3110	MAISON DE DEBAUCHE
3115	PROST-18,VIVRE PROD PROST
3120	PROXENETISME
3125	PROST-18,PROXENETISME
3130	AUTRE ACTE PROSTITUTION

Basic statistics

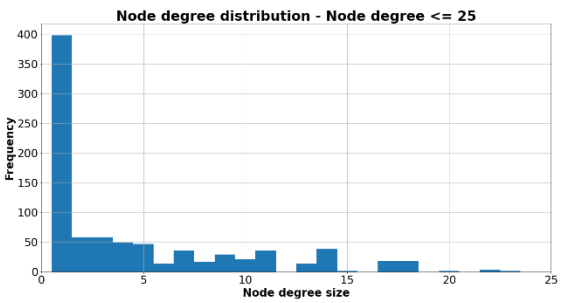
```
Number of cases after removing duplicates: 5404
Number of unique criminals: 3703
Number of unique crimes: 4779

Number of nodes: 3703
Number of isolated nodes: 2842
Number of edges: 2004

Mean of node degrees: 4.655
Number of nodes with degree >= 10: 154
Number of connected components: 250
```

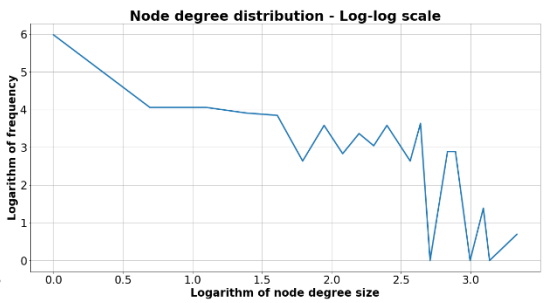
Crime Year	Nb of Crimes
2003	109
2004	136
2005	1147
2006	1358
2007	1040
2008	522
2009	467

We observe:

- 861 out of the 3703 nodes have at least one connection. We will thereafter only look at them (as we remove isolated components).
- 154 nodes are strongly connected (degree >= 10).
- A strong increase of crime cases in 2005 and 2006.

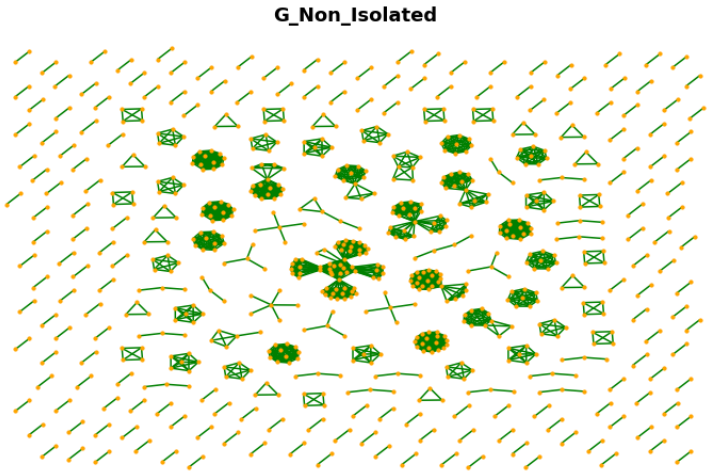
Degree distribution





Surprisingly, degree distribution is almost flat between k=2 and k=20. This is somewhat different from other studied networks, including the CAVIAR drug ring.

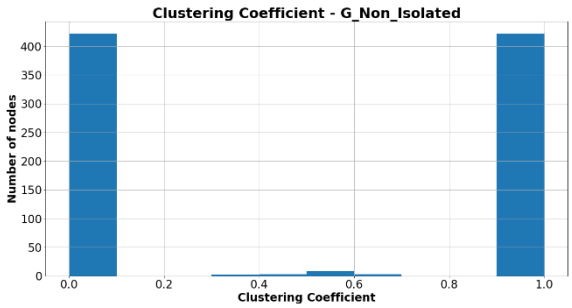
Visualisation



This representation of all non-isolated components reveals some interesting phenomenon:

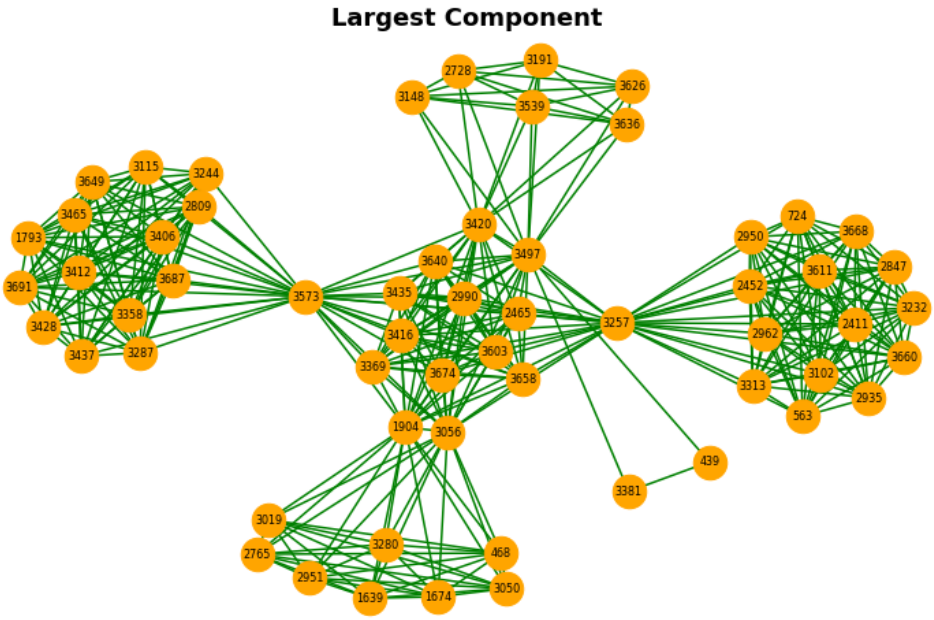
- **Many of the components are cluster graphs** (made of cliques)
- For the other ones (mostly the large components), a single node often links several cliques: these nodes with high betweenness centrality presumably play an important role in the organisation of the networks.

This explains the distribution of the clustering coefficient:



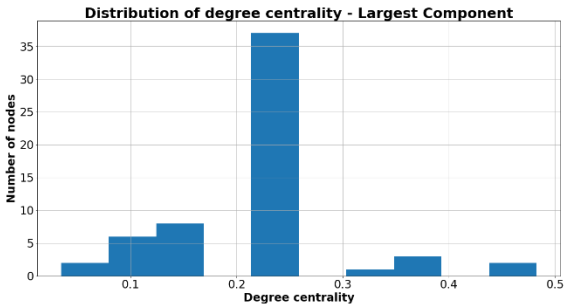
Focus on largest component

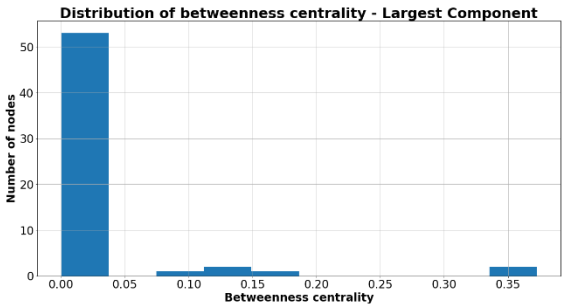
```
Number of nodes: 59
Number of edges: 389
Average degree:  13.186
```

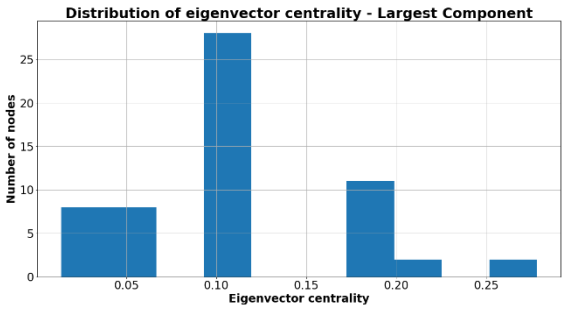


The shape of this network is very interesting, and strongly differs from the CAVIAR network:

- The five main clusters (~hairballs) are connected by one or two individuals (who happen to be women): obviously the few nodes with high betweenness centrality.







3. Discussion

Discoveries / surprises

- Contrarily to the CAVIAR network, the criminal networks studied here are mostly made of **cliques**.
- On **larger components**, it happens that a **few main clusters are connected by one (or a few) individual(s).**
- On **smaller components (cliques)**, finding the mastermind of the network is more challenging in the **absence of nodes with high betweenness centrality**.

Limitations of the study

- **Missing data / information**: we observe only the individuals that were arrested. Hidden links are probably missing.
- **Social Network Analysis (SNA) must be completed with qualitative / expert judgment** in order to understand who organizes the different networks.
- **Centrality measure** alone doesn't tell who organizes crime.